

CMSC 411 Computer Architecture, Spring 2025

Computer Science and Electrical Engineering
University of Maryland, Baltimore County

revision: January 28, 2025

1 Course Information

Instructor	Prof. Dong Li, 355 ITE Building, dli@umbc.edu
TA	Sina Makhdoomi Kaviri, sinam1@umbc.edu
Grader	Vishal Goud Sakkari, vsakkar1@umbc.edu
Lectures	229 ITE Building, Tuesday and Thursday, 02:30–03:45 PM
Office Hours	355 ITE Building, Tuesday, 10:00–11:00 AM or by appointment via email
Prereqs	Either <i>CMSC 313</i> or <i>CMPE 212</i> & <i>CMPE 310</i> with a grade of ‘C’ or better.

2 Description

This course covers the design of complex computer systems making heavy use of the components and techniques discussed in CMSC 313, or CMPE 212 and CMPE 310. All parts of the computer system—CPU, memory and input/output—are discussed in detail. Topics include information representation, floating-point arithmetic, instructions set design issues (RISC vs. CISC), microprogrammed control, hardwired control, pipelining, memory caches, bus control and timing, input/output mechanism and issues in the construction of parallel processors.

3 Objective

This course serves as a cornerstone in computer science education, bringing together foundational concepts from across the CSEE curriculum, including digital logic design, computer organization, system-level software, and engineering design. The course equips students with the knowledge and skills required for careers in computer science, offering a solid foundation for tackling real-world challenges and preparing them for advanced studies in graduate-level courses. It also provides a comprehensive framework for students interested in performance optimization, compilers, operating systems, and emerging technologies in hardware and software design. In this course, students will:

- Understand the fundamental principles and concepts of computer architecture, including instruction set architecture, memory organization, and pipelining.
- Analyze and evaluate the performance of computer systems through the use of metrics, benchmarks, and performance modeling techniques.
- Gain practical knowledge of designing and implementing various computer components, such as processors, memory systems, and input/output devices.

- Explore different types of parallelism, including instruction-level parallelism and thread-level parallelism, and understand their impact on computer system performance.
- Examine the role of caches and their organization in enhancing memory hierarchy and improving overall system performance.
- Study the principles of pipelining and its challenges, including hazards, dependencies, and techniques for mitigating them.
- Develop critical thinking and problem-solving skills by analyzing real-world case studies, current trends, and emerging technologies in computer architecture.

4 Communication

I encourage all students to actively participate in classroom discussions, whether by asking questions or sharing your perspectives. Your feedback throughout the semester is valuable to me, and I welcome it as we progress through the course.

- Course announcements, materials, and assignment submissions will be managed through **Blackboard**.
- For general inquiries or assistance, please use the **Blackboard discussion board or email**. When emailing, please include the following prefix in the subject line: [CMSC 411 Computer Architecture] <your subject>.
- Response time for emails is typically within 24 hours on weekdays.

5 Textbook

Computer Organization and Design: The Hardware/Software Interface: ARM Edition
(*The Morgan Kaufmann Series in Computer Architecture and Design*)

- **Author(s):** David A. Patterson, John L. Hennessy
- **ISBN#(s):** **ISBN-13:** 978-0128017333 and **ISBN-10:** 0128017333
- **Publisher:** Morgan Kaufmann
- **Edition:** 1st Edition (March 16, 2016)
- **Required or Recommended?** Strongly Suggested

6 Class Grading

The course grading is based on multiple components designed to evaluate students comprehensively. Class participation accounts for 5% of the total grade, emphasizing the importance of active engagement during the course. Homework constitutes a significant portion at 30%, ensuring students practice and understand the material. The remaining grades are distributed across exams: the First Exam and Second Exam each contribute 20%, assessing students' understanding of key topics throughout the semester. Finally, the Final Exam makes up 25% of the grade, testing students' cumulative knowledge of the course material. This distribution ensures a balanced evaluation of participation, consistent effort, and academic performance.

Description	Grade distribution
Class Participation	5%
Homework	30%
First Exam	20%
Second Exam	20%
Final Exam	25%

- **Class Participation:** Participation is more important than mere attendance, and active engagement in class discussions and activities is highly encouraged. While attendance will be tracked using Poll Everywhere to support the participation grade, if you are unable to attend a class, please email your reason beforehand to ensure transparency and allow for necessary support. Lecture notes will remain accessible throughout the semester. While I am committed to being present for every class, I expect the same from you. If you miss a class, I am not responsible for reteaching the material, and office hours are not intended for recapping missed lectures. Instead, connect with classmates to share notes and insights, and review the provided lecture slides. Always notify me prior to class if you anticipate being absent, explaining the reason for your absence.
- **Homework:** Homework assignments are designed to take approximately 30 minutes to complete and will be made available at 8:30 PM on the day of the assignment following the last section's lecture. Students will have one week to complete their homework, providing two class sessions to ask questions, and must submit their work as a PDF on Blackboard by 11:59 PM on the due date. Graders aim to return feedback within one week. The Grader has full authority over homework-related matters, and students should direct any questions to the Grader, with the instructor stepping in only to resolve significant issues. At the end of the semester, it is the student's responsibility to ensure all homework grades are accurate, as no grade reviews will occur after the final class. Late submissions will incur a penalty of 10% per class day, with a maximum deduction of up to 50%.
- **Exams:** The course assessments will consist of cumulative tests, including First Exam, Second Exam, and Final Exam, all conducted through Blackboard. All exams will be open book and open notes, allowing students to reference course materials during the exams. This ensures that the focus remains on applying knowledge rather than solely on memorization.

Grade	Percentage Range
A	90% or better
B	80% to 90%
C	70% to 80%
D	60% to 70%
F	Below 60%

Letter grades in this course will be assigned based on the following percentage ranges: an A represents 90% or better, a B is 80% to 90%, a C is 70% to 80%, and a D is 60% to 70%. An F indicates performance below 60%. For students enrolled in the Pass/Fail option, a grade of 70% or higher is required to earn a Pass.

7 Schedule

The class schedule is tentative and subject to change.

#	Week	Date	Topic	HMWK Assigned	HMWK Due	HMWK Graded
1	1	Tue, Jan 28, 2025	Introduction and Overview	HW#1		
2		Thu, Jan 30, 2025	Performance Evaluation and Metrics			
3	2	Tue, Feb 04, 2025	Logic and Making Ics	HW#2	HW#1	
4		Thu, Feb 06, 2025	Instruction Set Architecture - ARM			
5	3	Tue, Feb 11, 2025	Instruction Set Architecture - ARM/MIPS/x86	HW#3	HW#2	HW#1
6		Thu, Feb 13, 2025	Arithmetic/Carry/Floating Point #1			
7	4	Tue, Feb 18, 2025	Arithmetic/Carry/Floating Point #2	HW#4	HW#3	HW#2
8		Thu, Feb 20, 2025	CPU			
9	5	Tue, Feb 25, 2025	Review	HW#5	HW#4	HW#3
10		Thu, Feb 27, 2025	First Exam (Blackboard)			
11	6	Tue, Mar 04, 2025	Pipelining 1	HW#6	HW#5	HW#4
12		Thu, Mar 06, 2025	Pipelining 2 + First Exam Review			
13	7	Tue, Mar 11, 2025	Pipelining 3	HW#7	HW#6	HW#5
14		Thu, Mar 13, 2025	Memory			
15	8	Tue, Mar 18, 2025	Spring Break		HW#7	HW#6
16		Thu, Mar 20, 2025	Spring Break			
17	9	Tue, Mar 25, 2025	Cache #1	HW#8		HW#7
18		Thu, Mar 27, 2025	Cache #2			
19	10	Tue, Apr 01, 2025	Cache Performance	HW#9	HW#8	
20		Thu, Apr 03, 2025	Virtual Memory			
21	11	Tue, Apr 08, 2025	Review	HW#10	HW#9	HW#8
22		Thu, Apr 10, 2025	I/O Systems			
23	12	Tue, Apr 15, 2025	Second Exam (Blackboard)	HW#11	HW#10	HW#9
24		Thu, Apr 17, 2025	Superscalar			
25	13	Tue, Apr 22, 2025	VHDL + Second Exam Review	HW#12	HW#11	HW#10
26		Thu, Apr 24, 2025	Superscalar			
27	14	Tue, Apr 29, 2025	Embedded		HW#12	HW#11
28		Thu, May 01, 2025	Microprogramming			
29	15	Tue, May 06, 2025	Supercomputers			HW#12
30		Thu, May 08, 2025	Optimizing Code & Algorithms			
31	16	Tue, May 13, 2025	Final Review			
32	17	Tue, May 20, 2025	Final Exam (Blackboard)			

Figure 1: Class Schedule.

8 Class Policies

- **Academic Integrity**

Plagiarism or any form of cheating is not tolerated in this class. All submitted work must be your own. If you are suspected of cheating, plagiarism, or other forms of academic dishonesty, your case will be referred to the Undergraduate Academic Conduct Committee or the Graduate Council Grievance Committee, and this may result in lost credit, course failure, suspension, or dismissal from UMBC, depending on the Committee's decision. For more information on what constitutes academic dishonesty, please consult UMBC Academic Conduct. If you are struggling with the class content in any way, it is your responsibility to communicate with the instructor so that they can assist you.

- **Generative AI Tools**

You are open to use GenAI tools. But you need to mention the specific tasks where genAI tools were used and the steps you took to ensure accuracy and quality.

9 UMBC School Policies

- **Accessibility and Disability Accommodations, Guidance and Resources**

Accommodations for students with disabilities are provided for all students with a qualified disability under the Americans with Disabilities Act (ADA & ADAAA) and Section 504 of the Rehabilitation Act who request and are eligible for accommodations. The Office of Student Disability Services (SDS) is the UMBC department designated to coordinate accommodations that create equal access for students when barriers to participation exist in University courses, programs, or activities.

If you have a documented disability and need to request academic accommodations in your courses, please refer to the SDS website at sds.umbc.edu for registration information and office procedures.

SDS email: disAbility@umbc.edu

SDS phone: 410-455-2459

If you will be using SDS approved accommodations in this class, please contact the instructor to discuss implementation of the accommodations. During remote instruction requirements due to COVID, communication and flexibility will be essential for success.

- **Sexual Assault, Sexual Harassment, and Gender-Based Violence and Discrimination**

UMBC Policy in addition to federal and state law (to include Title IX) prohibits discrimination and harassment on the basis of sex, sexual orientation, and gender identity in University programs and activities. Any student who is impacted by sexual harassment, sexual assault, domestic violence, dating violence, stalking, sexual exploitation, gender discrimination, pregnancy discrimination, gender-based harassment, or related retaliation should contact the University's Title IX Coordinator to make a report and/or access support and resources. The Title IX Coordinator can be reached at titleixcoordinator@umbc.edu or 410-455-1717.

You can access support and resources even if you do not want to take any further action. You will not be forced to file a formal complaint or police report. Please be aware that the University may take action on its own if essential to protect the safety of the community.

If you are interested in making a report, please use the Online Reporting/Referral Form. Please note that, if you report anonymously, the University's ability to respond will be limited.

Notice that Faculty and Teaching Assistants are Responsible Employees with Mandatory Reporting Obligations.

All faculty members and teaching assistants are considered Responsible Employees, per UMBC's Policy on Sexual Misconduct, Sexual Harassment, and Gender Discrimination. Faculty and teaching assistants are therefore required to report all known information regarding alleged conduct that may be a violation of the Policy to the Title IX Coordinator, even if a student discloses an experience that occurred before attending UMBC and/or an incident that only involves people not affiliated with UMBC. Reports are required regardless of the amount of detail provided and even in instances where support has already been offered or received.

While faculty members want to encourage you to share information related to your life experiences through discussion and written work, students should understand that faculty are required to report past and present sexual harassment, sexual assault, domestic and dating violence, stalking, and gender discrimination that is shared with them to the Title IX Coordinator so that the University can inform students of their rights, resources, and support. While you are encouraged to do so, you are not obligated to respond to outreach conducted *as a result of* a report to the Title IX Coordinator.

If you need to speak with someone in confidence, who does not have an obligation to report to the Title IX Coordinator, UMBC has a number of Confidential Resources available to support you:

– **Retriever Integrated Health (Main Campus):**

Phone: 410-455-2472; Monday – Friday 8:30 a.m. – 5 p.m.

For After-Hours Support, Call 988.

– **Center for Counseling and Well-Being (Shady Grove Campus):**

Phone: 301-738-6273; Monday–Thursday 10:00 a.m. – 7:00 p.m.

Friday 10:00 a.m. – 2:00 p.m. (virtual)

Online Appointment Request Form.

– **Pastoral Counseling via The Gathering Space for Spiritual Well-Being:**

Phone: 410-455-6795; Email: i3b@umbc.edu; Monday – Friday 8:00 a.m. – 10:00 p.m.

– **Women’s Center (open to students of all genders):**

Phone: 410-455-2714; Email: womenscenter@umbc.edu

Monday – Thursday 9:30 a.m. – 5:00 p.m. and Friday 10:00 a.m. – 4:00 p.m.

– **Child Abuse and Neglect**

Please note that Maryland law and UMBC policy require that faculty report all disclosures or suspicions of child abuse or neglect to the Department of Social Services and/or the police even if the person who experienced the abuse or neglect is now over 18.

– **Additional Resources:**

Shady Grove Student Resources, Maryland Resources, National Resources.

• **Pregnant and Parenting Students**

UMBC’s Policy on Sexual Misconduct, Sexual Harassment and Gender Discrimination expressly prohibits all forms of discrimination and harassment on the basis of sex, including pregnancy. Resources for pregnant, parenting, and breastfeeding students are available through the University’s Office of Equity and Civil Rights. Pregnant and parenting students are encouraged to contact the Title IX Coordinator to discuss plans and ensure ongoing access to their academic program with respect to a leave of absence—returning following leave, or any other accommodation that may be needed related to pregnancy, childbirth, adoption, breastfeeding, and/or the early months of parenting.

In addition, students who are pregnant and have an impairment related to their pregnancy that qualifies as a disability under the ADA may be entitled to accommodations through the Office of Student Disability Services.

• **Religious Observances & Accommodations**

UMBC Policy provides that students should not be penalized because of observances of their religious beliefs, and that students shall be given an opportunity, whenever feasible, to make

up within a reasonable time any academic assignment that is missed due to individual participation in religious observances. It is the responsibility of the student to inform the instructor of any intended absences or requested modifications for religious observances in advance, and as early as possible. For questions or guidance regarding religious observances and accommodations, please contact the Office of Equity and Civil Rights at ecr@umbc.edu.

- **Hate, Bias, Discrimination, and Harassment**

UMBC values safety, cultural and ethnic diversity, social responsibility, lifelong learning, equity, and civic engagement.

Consistent with these principles, UMBC Policy prohibits discrimination and harassment in its educational programs and activities or with respect to employment terms and conditions based on race, creed, color, religion, sex, gender, pregnancy, ancestry, age, gender identity or expression, national origin, veterans status, marital status, sexual orientation, physical or mental disability, or genetic information.

Students (and faculty and staff) who experience discrimination, harassment, hate, or bias based upon a protected status or who have such matters reported to them should use the online reporting/referral form to report discrimination, hate, or bias incidents. You may report incidents that happen to you anonymously. Please note that, if you report anonymously, the University's ability to respond may be limited.